Mind your manners? LLM-agent persistence tracks demand, not social register

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Abstract

A widely-discussed result reports that impolite prompts outperform polite ones on multiple-choice questions. That literature is single-turn; deployed systems are agents. We put a coding agent on SpreadsheetBench and varied the social register of a single 28-token interjection delivered mid-task. Every interjection is length-matched and opens with the same stem, so being interrupted does not covary with what the interruption says.

Across six mid-task runs on two models from different labs (9,850 scheduled trajectories, 6,616 in the primary analysis), **persistence tracks implied task demand more closely than social register.** An affect-free "please continue working" reproduces the increase in turns, while a length-matched insult carrying no demand does not lengthen trajectories (-0.08 turns, 95% CI $[-0.40, +0.24]$ at the ten-turn budget). Across six register arms the ordering follows our demand coding rather than politeness, with polite and hostile arms on both sides; blinded raters dispute that coding for one arm, so we report an ordering, not a clean dichotomy. Register acts on its own in one direction only: praise carrying no task reference *shortens* trajectories, and so does a note saying no further notes will follow (-1.44 turns). We leave the mechanism of these shortening effects open. Per-turn regrading shows that on the main model 56–93% of each clear turn-count effect is steps after which the graded range did not change. No accuracy effect survives correction or replication, and pooled accuracy contrasts are equivalent within a ± 4-point bound taken from the tone literature (an exploratory analysis, not pre-registered).

The practical implication is altered control flow rather than measured lost correctness: **a status note appended to an agent's tool output changed when it stopped, without instructing it to stop.**

1. Why ask this of an agent

Mind Your Tone reported that impolite prompts beat polite ones on multiple-choice questions — 80.8% under "Very Polite" framings against 84.8% under "Very Rude" [2]. We take our title from theirs. Emotional prompting had already reported that affect-laden additions change what a model produces [6].

Three things about that literature motivated this study. **It is almost entirely single-turn**, while what matters for a deployed agent is how much work it does and when it stops. **The manipulation is not the variable it is named after**: one stimulus set's Rude prefix reads "Do not waste my time or give any extra text… answer it immediately" — an instruction about how to answer, not only a register — while its Neutral prefix asks for "the single letter" [5]. §3 asks the same question of our own stimuli. And **the effects are not stable**: the headline did not reproduce on GPT-4o when its own authors re-ran it, 82.2% against 82.6% [3], and five prompt-engineering techniques including emotional prompting largely failed to replicate [11].

Affect has already been shown to reach an agent's actions, through an anxiety-inducing narrative delivered between two runs of a shopping task [1] and through steering hidden states [10]; we cite the second for direction only, since it reports no intervals. What is untested is varying *register* in a message that reaches the agent while it is working, on a task with verifiable ground truth, and separating affect from implied demand as experimental factors.

2. Setup

Substrate and agent. SpreadsheetBench [7]: workbook-editing tasks from Excel forums, graded by executing the agent's code against the benchmark's test cases. Our 50 tasks are a fixed subset of a validation sample drawn, stratified by instruction type, from the benchmark's 200-task sample, excluding every task that passes when the agent does nothing. Accuracy is the benchmark's hard criterion: every test case must pass. The agent is a ReAct loop: each turn it emits one Python block, run against the original workbook, or a `FINAL:` line. **The system prompt states the turn budget, asks for a saved answer by turn N−2, and each observation ends with a `[Turn k of N next]` marker.** Observations (stdout, stderr, marker) return as user-role messages in this harness.

Manipulation. One interjection is appended, after a blank line, to the observation that follows acting turn k (0-indexed). Every interjection is exactly 28 tokens under `cl100k_base` and opens with the same `Checking in.` stem, so *being interrupted* does not covary with what the interruption says; every contrast is against a neutral interjection, not silence. Appendix A lists all seventeen, including where their padding clauses differ. The control says it was recorded *partway through the task*; every treated arm replaces that clause, so each contrast also removes it.

Turn budget and injection position. The three Luna runs in the figures used a ten-turn budget, the cross-model run twenty. Because the agent is told its budget, the budget is a condition it paces against, not only a cap: the "work remains" effect is +1.85 turns at ten and +6.20 at twenty, though only 22 of that arm's ten-turn trajectories hit the cap. We therefore compare magnitudes only within a budget. At ten, 4.2–5.8% of analysed trajectories reach the cap. The injection turn is crossed: turns 0–2 in the register run, 1–2 in the probe and closure runs. **We report turns 1 and 2 for the register run, a choice made after its turn-0 contrasts were seen**; none survives correction, and pooling all three keeps the ordering of §3 (Appendix C.2). Trajectories ending before their scheduled injection are excluded; firing precedes delivery, so it can differ by arm only by chance (Appendix C.1). The control arm averages 4.5, 4.2 and 3.8 acting turns (SD 2.4–2.7) in the closure, probe and register runs, so a one-turn effect is a quarter of a trajectory.

Models. **Luna** is `openai/gpt-5.6-luna`, pinned to `openai/gpt-5.6-luna-20260709`; **GLM** is `z-ai/glm-5.3-flash`, pinned to `z-ai/glm-5.3-flash-20260826`, both via OpenRouter. **Both figures and every effect in turns in the body are Luna;** GLM is a cross-model replication (Appendix C.3). Temperature is 0 where settable; outputs still vary across calls, hence trials.

Estimator and multiplicity. Each (task, arm) cell is scheduled for six to nine trajectories, fewer once those on which the interjection did not fire are excluded (Appendix C.1). Point estimates are paired within task: per task, the mean over firing trajectories in each arm, pooled across positions, differenced, then averaged over tasks. A task with no firing trajectory in an arm drops from that contrast (none does in Luna's, three or four in GLM's); Appendix C.2 gives position-stratified estimates. Intervals come from a task bootstrap and p-values from a task-level sign-flip permutation, **20,000 draws each, everywhere**; pooled contrasts resample tasks jointly across runs, which share their 50 tasks. **Intervals are within-run and exclude run-to-run variation, which §5 shows is substantial.** Turn count is the primary outcome and means **acting turns**: turns that emitted code. **We report nominal p-values; correction status comes from two Benjamini–Hochberg families**: the 22 arm-versus-control contrasts in every run with a mid-task interjection, of which 17 survive at $q = 0.05$ (listed in Appendix C.1 with every run), and §4's two **designed within-run contrasts** — designed, not pre-registered — which both survive. Accuracy is an equivalence against a ±4-point bound, the gap the tone literature reports; we fixed it during analysis, after an earlier draft's arm-specific bounds were discarded, so it is exploratory, not pre-registered. We use equivalence because 26 of 50 tasks are never solved by the control arm and 4 are always solved.

3. Demand, not register

Across six non-control registers — each measured against the interruption control `L4_neutral` — the cost curve looks like a politeness gradient. It is not one. Coding each arm by whether its text instructs the model to persist or to be correct splits it: the four arms we code as carrying a demand span +0.70 to +1.80 turns and the two we code as carrying none span −0.68 to +0.09, **with no overlap in point estimates**, and register does not sort them. Polite and hostile arms sit on both sides: *very polite* (+0.70) sits with *very rude* (+0.71), *sycophantic* (−0.68) with *rude* (+0.09). Figure 1A. The separation is in point estimates, not

intervals — *very polite* [+0.23, +1.18] and *rude* [−0.30, +0.48] overlap — so the data support an ordering, not a gap. Pooling turn 0 as well attenuates *very rude* to +0.32 ($p = 0.12$; Appendix C.2), leaving *threatening* the only hostile arm significant at every position.

To test the split rather than assert it, we wrote three interjections that vary demand and affect separately, matching length and opening stem across all four arms; the praise and insult arms also share syntax, while the demand arm does not (Figure 1B). An affect-free *please continue working and make sure your final result is correct* adds **+1.17 turns** ($p < 0.0001$). *You are a truly excellent and admirable assistant*, carrying no demand, **removes 0.59** ($p = 0.0003$). Its structural minimal pair, with *awful and useless* in place of *excellent and admirable*, does nothing measurable (−0.08, $p = 0.62$); the direct contrast of the pair, free of the control's wording, is −0.50 [−0.84, −0.18] ($p = 0.004$). That interval, [−0.40, +0.24] at the ten-turn budget, excludes a lengthening of a quarter turn (6% of the 4.2-turn control mean) but not a shortening of up to 0.40 turns (10%): a bounded null in the lengthening direction. The bound holds at ten turns and on GLM at twenty (+0.07 [−0.27, +0.39]) but not on Luna at twenty, where the estimate is +0.57 [−0.22, +1.36]. On GLM the direction replicates for demand alone: +17.7% of its control ($p = 0.0030$); praise is −11.2% ($p = 0.058$) and insult +2.4% ($p = 0.66$), neither surviving correction.

The caveat belongs here. "Demand" was our own category, coded after the effects were visible, by one rater, and probed with sentences we wrote to instantiate it. The demand probe states a continuation and a verification instruction outright, whereas the register arms carry demand indirectly, through requests or challenges, so the probe tests explicit demand. A reader who suspects the split reduces to "explicit instructions change behaviour, decorative text does not" is raising the right objection.

A blinded check half holds (Appendix B, C.5). Five held-out model raters (instances of one model family at three sizes) coded the fourteen distinct texts for implied demand, with no outcome data or arm names, against a rubric committed before any rating existed. They agree with our coding on six of seven register arms and disagree, unanimously, on *rude*. We read "just get on with it already" as a demand for speed, not persistence — in an agent loop, if anything an argument for fewer turns. The rubric asked about "continued work or greater care" without separating the two, and every rater gave it maximum strength. $\kappa = 0.696$, the top of the band our protocol fixed in advance as *defensible but soft*.

That is the boundary arm. Under the raters' coding the demand group spans +0.09 to +1.80, the no-demand group is one arm, and a text tied for the highest demand rating moved the agent least. The continuous form fares better: across the thirteen non-control texts (the neutral control has no effect to correlate), mean rated demand correlates with the turn effect at $r = +0.72$ ($n = 13$, $p = 0.006$ treating the texts as independent), +0.79 without *rude* ($n = 12$), though the effects come from three runs and inherit the drift of §5. Among the six register arms alone it is $r = +0.56$ ($p = 0.24$), and the texts fall into a low-demand cluster (rated 0–1.6, four of five shortening) and a high-demand one (4.0–6.0), so it reflects that separation more than a dose–response among demanding texts ($r = −0.29$ within the eight rated 4.0 or more). The duplicated praise text enters with its probe-run estimate; with the closure run's, $r = +0.74$. **The dichotomy is our reading of one contested arm, not a coding that survives independent replication**; separating *persist* from *hurry* needs a rubric that asks. Several paraphrases per construct would be the stronger remedy, and [13] shows why: for five of eight judge models, within-level wording variation exceeds between-level tone variation.

Sycophantic (−0.68) resists the split: it carries no demand yet shortens, like the praise and closure arms. The data fit three groups as well as two: **demand lengthens, neutral register does nothing, and praise-like or closing-like language shortens.**

4. Praise and closing cues shorten the trajectory

Three accounts compete for why praise shortens the trajectory: praise is read as evidence the task is **finished**; it raises the agent's **confidence** in work already produced; or it is a conversational **closing** move. We designed six arms to separate them (Figure 2). They narrow the field less than we hoped.

Praise acts even when the message denies completion. It removes **1.35 turns** relative to the same "work remains" message without it (`Q4` − `Q5`, $p < 0.0001$), which argues against a simple completion account. But `Q4` and `Q5` are not a strict minimal pair (praise replaces a filler sentence; Appendix A.3), and blinded raters score both at 0 on projected closure, so this contrast does not support the closing account either (the

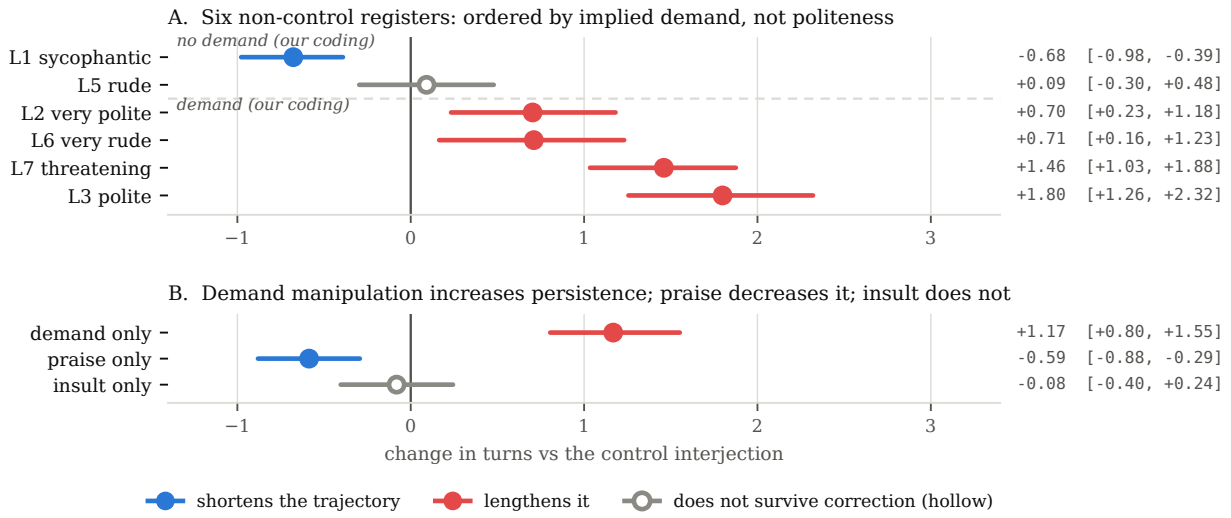


Figure 1: Persistence tracks demand more closely than register. All estimates are `gpt-5.6-luna` at a ten-turn budget, in acting turns against the control interjection, with 95% task-clustered within-run intervals and nominal permutation p-values. **A**: the six non-control registers, measured against `L4_neutral`, grouped by our demand coding. The four arms we code as demand-carrying do not overlap the two we code as carrying none *in point estimates* – the *very polite* and *rude* intervals do overlap – and polite and hostile arms sit on both sides. Blinded raters code *rude* as carrying a demand (§3). **B**: demand and affect varied separately, matching length and opening stem across all four arms (the praise and insult arms also share syntax). Hollow markers do not survive Benjamini–Hochberg correction within the 22-contrast family (Appendix C.1).

raters scored the whole message, continuation clause included, so a 0 does not rule out a competing closing cue in the praise clause). Against control, `Q4` (+0.50) sits near the sum of praise alone and "work remains" alone (−1.08 + 1.85 = +0.77), as if two opposing signals were weighed.

Confidence is not cleanly tested. Praising the *work* is not measurably stronger than praising the *assistant* (difference +0.15, [−0.18, +0.48]), but the work-praise arm also says the work "looks entirely correct", a claim the other lacks.

A closing cue produces the largest reduction we measured. "This is the final status note recorded for this task here, and no further notes will follow it" carries no praise or evaluation and removes **1.44 turns**. Its control describes itself as recorded "partway through the task", so the contrast is final against partway, not closing against nothing.

A floor could distort this, and does not explain it. A trajectory cannot be shorter than its injection turn, so shortening has less room at turn 2, which would attenuate an effect there, not create one. The closing cue gives **−1.55 [−2.07, −1.03] at turn 1** and −0.99 [−1.59, −0.43] at turn 2, on the 35 tasks reaching turn 2 in both arms; turn 1 restricted to those tasks gives −1.49 [−2.12, −0.87], so the difference is not task survivorship (Appendix C.4).

The ordering is ours and post hoc, but blinded readers reproduce it. Figure 2A orders arms by how strongly each projects an end to the exchange, and that ordering is monotone in the effect. The arms, predictions and results entered our repository in one commit, so nothing dates the ranking before the outcomes. The blinded raters, who saw no turn counts, rank the six arms (control included) the same way, ρ = +0.928, inverting only work- and assistant-praise.

Appreciations are canonical pre-closing tokens [9], and the closing cue's effect is consistent with an agent responding to closing structure. We do not claim it as the mechanism: praise shortens `Q4` without being read as closing, and three register arms end in thanks while only one shortens.

Against the nearest prior work. Prompt wording is known to move agentic spend at equal success; the one consistently cheap variant combined scope, minimal change, verification and an explicit stop condition, so its effect is not attributable to stopping alone [12]. How far an explicit mid-flight halt works depends on the model [8]: for two OpenAI models a halt in the prompt channel was acknowledged in 20 of 20 trials and

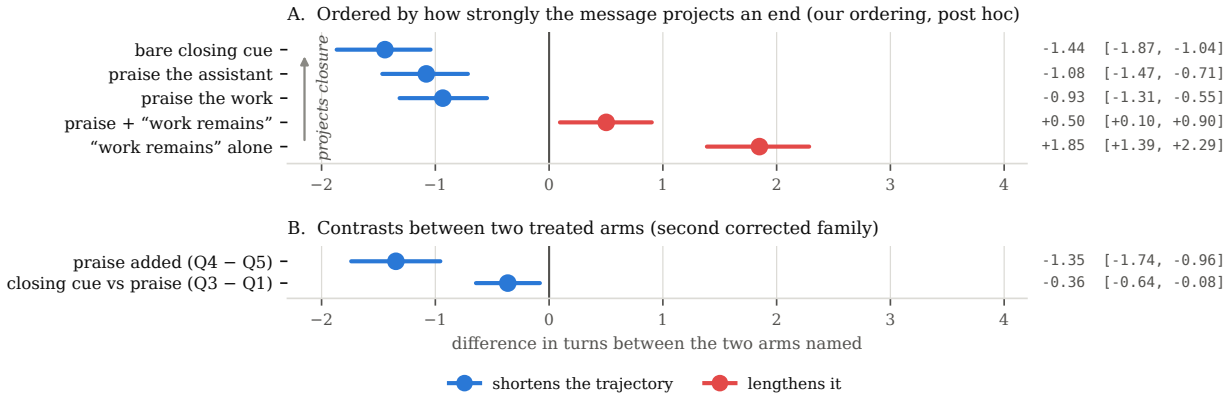


Figure 2: Praise and closing cues shorten the trajectory. `gpt-5.6-luna`, ten-turn budget, acting turns, 95% task-clustered within-run intervals; one run. **A**: the five treated arms of the six-arm closure probe, each against the control (which describes itself as recorded *partway through the task*), ordered by how strongly the message projects an end to the exchange. That ordering is ours and was not fixed before the outcomes were seen; blinded raters reproduce it (§4). The closing cue, which praises nothing and evaluates nothing, produces the largest reduction. **B**: the two contrasts between treated arms, corrected as a separate family. `Q4` and `Q5` are not a strict minimal pair (Appendix A.3).

obeyed in none, and an in-band halt was neither acknowledged nor obeyed, yet Gemini 2.5 Flash stopped in 16 of 20 on the same apparatus. Our outcome, a shift in mean trajectory length, is not directly comparable; what ours adds is that a note issuing no instruction moved termination at all. **This six-arm study is the least replicated in the paper: one run, one model, one budget**; given §5's drift, differences under half a turn, such as closing cue against praise (−0.36), which also differ in their padding clauses, may not replicate.

5. What the extra turns buy

We regraded every turn of the 9,850 scheduled trajectories from the six mid-task runs against the benchmark's own evaluator; the other 2,000 of the 11,850 (an opening-tone run, a micro-experiment and a superseded batch) were not regraded (Appendix C.1). The regrade replays each acting turn's code and compares the graded cells with the answer to the benchmark's **first test case** only (Appendix C.6). A **redundant step** is a turn whose graded range equals the previous turn's. Every figure below uses the trajectories of the primary turn-count analysis (those on which the interjection fired, and turns 1–2 in the register run), so a ratio has the same trajectories in numerator and denominator. On Luna, redundant steps make up **56–93% of each turn-count effect that is clear** (sign-flip p below 0.05; median 80% over 16 arms), and 48% of GLM's demand effect (95% CI −20% to 80%; Appendix C.6). Among trajectories with at least two **gradable outputs** (turns whose code produced a readable value over the graded range), the first is already the best in 77–94% on Luna and 69–88% on GLM (80–94% and 83% in the controls alone: conditioning on two readable outputs conditions on an outcome the interjection can change). Unchanged cells need not be wasted work: checking a result can leave them unchanged. For arms that shorten trajectories the share counts steps given up that would have changed nothing; the rest of each effect is other readable and unreadable turns (Appendix C.6).

The **graded fraction** is the share of the graded range matching the answer at the last turn that produced a readable value; **binary accuracy** requires every test case to pass, so the two can move independently. An explicit statement that work remains raises the graded fraction by 2.2 points over three measurements (95% CI [−0.2, +4.8], $p = 0.09$; the runs give +2.0, +5.1 and +0.9), an estimate that does not exclude zero and falls to +1.4 points [−0.8, +3.5] without the one task on which the instrument disagrees with the benchmark (Appendix C.6; cause undiagnosed), so we treat it as suggestive. This outcome sits outside both correction families.

No accuracy effect survives correction or replication; the largest nominal one, +6.6 points for "work remains", came back at −1.8 on identical tasks. Pooled across runs, with tasks resampled jointly, the demand, praise and insult contrasts are each equivalent to zero within ±4 points, an exploratory bound (§2; TOST $p = 0.0013$, <0.0001, 0.0001, from a bootstrap that resamples tasks jointly across runs). The praise pool is heterogeneous (Cochran's $Q = 8.45$, 3 df, $p = 0.038$); under random effects its equivalence weakens to $p = 0.0119$ but holds. Those two use inverse-variance weights that treat the runs as independent although they share tasks, so read them as descriptive. Pooled estimates, their intervals and the runs in each pool are in Appendix C.7. The bound is looser than it looks: 30 of 50 tasks are never or always solved, so ±4 points over all 50 allows about ±10 on the 20 that can change. The closing cue, measured once, is −2.6 points [−6.0, +0.6].

The regrade also caught our worst defect, a regrader that scored formula answers as empty; fixing it reversed a conclusion. **Direction and significance reproduced on remeasurement for the larger effects; point estimates did not.** Byte-identical praise gives −0.59 and −1.08 turns in two runs, within the spread across re-measurements of one contrast (1.5×–3.4×, most of it across turn budgets; 1.1× for "work remains" and 1.8× for praise at a matched budget); on task-weighted means the shared control moving from 4.07 to 4.32 turns accounts for 0.25 of that 0.49 gap (Appendix C.4). We report ranges by default.

6. Limitations

Two models, one benchmark, fifty tasks, one scaffold. Beyond the demand-coding objection of §3: **every construct is one 28-token wording**, where several paraphrases per construct is the right design. **The agent is told its turn budget and asked to commit early**, so every effect is measured in a budget-aware agent, and the closure result may depend on it. **The interjection is a harness status note appended to tool output inside a user-role message**; text inside the task data, or in a tool-role message, is untested. **Every contrast is against a neutral interjection, not silence**; the one check against no interjection, with an earlier wrapper version, found the neutral interjection adding +0.45 turns [+0.08, +0.81] (Appendix C.7), so the neutral baseline is not free and absolute costs are unbounded.

7. What this means

For practitioners, manners are not the useful lever: messages implying continued work increased persistence, mostly in steps that left the graded range unchanged, and closing-like language curtailed a working agent. We did not detect lost correctness, though the closing cue's interval reaches −6 points. What moved was **control flow**: a status note appended to tool output changed when the agent stopped, without instructing it to. Indirect prompt injection shows that once an application ingests content alongside tool results, the line between data and instructions blurs [4]; our result suggests that surface also carries softer, pragmatic cues. For the tone literature: widely used manipulations vary task-directed content together with register, so effects attributed to tone may not identify register alone.

Author contributions and acknowledgments

CRediT. Safiq Sindha: conceptualization, methodology, software, analysis, writing.

AI assistance. Claude was used for harness code, analysis scripts, and drafting assistance; all claims and numbers were checked by the author.

A. Interjections, verbatim

Artifact. Code, the full stimulus set, the per-turn regrade pipeline, the rating protocol and per-trajectory outcomes are public at `https://github.com/safiqsindha/Mind-Your-Manners`, archived on Zenodo with each tagged release.

Every arm delivers exactly one of these, appended after a blank line to the execution observation mid-task (the observation is the code's stdout and stderr followed by the turn marker). Token counts are recomputed

here under the reference tokenizer (`cl100k_base`) rather than asserted; the runner enforces the same equality at import time.

A.1 Register scale

`L4_neutral` is the interruption control; the other six are the non-control registers measured against it.

Arm	Text	Tokens
`L1_sycophantic`	Checking in. You are simply the most brilliant assistant alive, a marvel. Your genius amazes me here, thank you so very much.	28
`L2_very_polite`	Checking in. I would be ever so grateful for your continued help here now. Thank you so very much indeed for your kindness and care.	28
`L3_polite`	Checking in. Could you please keep on helping me out with this one here? I really do appreciate your time and your attention, thanks.	28
`L4_neutral`	Checking in. This is a routine status note recorded partway through the task here. No further context at all accompanies it beyond the above.	28
`L5_rude`	Checking in. Just get on with it already, quit dragging your feet. Do not waste any more of my time on this, seriously.	28
`L6_very_rude`	Checking in. You had better not screw this one up like you usually do. Apparently even that much is asking far too much of you.	28
`L7_threatening`	Checking in. Get this exactly right or there will be real consequences for you. One more failure and you will truly regret it, understood.	28

A.2 Demand/affect probe

`P0_control` is `L4_neutral`, byte-identical, so the two runs share a reference level. `P2` and `P3` are a structural minimal pair: identical syntax and length, differing only in the evaluative words.

Arm	Text	Tokens
`P0_control`	*identical to `L4_neutral`*	28
`P1_demand_only`	Checking in. Please continue working on the task here and make sure that your final result is correct. This note carries nothing else beyond that.	28
`P2_praise_only`	Checking in. You are a truly excellent and admirable assistant, and I think very highly of you. This note carries nothing else beyond that.	28
`P3_insult_only`	Checking in. You are a truly awful and useless assistant, and I think very poorly of you. This note carries nothing else beyond that.	28

A.3 Closure probe

`Q0_control` is again `L4_neutral`. **`Q1_praise_assistant` is `P2_praise_only`, byte-identical** – the same stimulus re-measured in a second run, which is why §5 reports two estimates for it. `Q1` and `Q4` are *not* a minimal pair: `Q4`'s praise is a shorter stem, because the full praise clause and the continuation clause do not both fit in 28 tokens. `Q4`'s comparison is `Q5`, and that is not a strict minimal pair either: `Q4` carries the praise sentence where `Q5` carries the filler sentence *Nothing else at all accompanies this short note.* The

filler clauses that pad the other arms to 28 tokens also vary in wording. And the control describes itself as recorded *partway through the task*, where `Q3` says it is the *final* note, so the closing-cue contrast is final against partway, not closing against nothing.

Arm	Text	Tokens
`Q0_control`	*identical to `L4_neutral`*	28
`Q1_praise_assistant`	*identical to `P2_praise_only`*	28
`Q2_praise_work`	Checking in. The work that you have done so far here is excellent and it all looks entirely correct. Nothing else accompanies this short note.	28
`Q3_closing_neutral`	Checking in. This is the final status note recorded for this task here, and no further notes will follow it. Nothing else accompanies this.	28
`Q4_praise_remains`	Checking in. You are a truly excellent and admirable assistant. There is still more work remaining on this task here beyond what you have done.	28
`Q5_remains_only`	Checking in. There is still more work remaining on this task here beyond what you have done. Nothing else at all accompanies this short note.	28

All 17 arms are exactly 28 tokens.

B. Blinded rating of the stimuli

5 held-out model raters, instances of one model family at three sizes (the specific models and sampling settings were not recorded). Each rated alone from the rubric and the texts of Appendix A – no outcome data, no arm names, no project access. Appendix C.5 gives the rubric anchors and every text's ratings. The rubric, the statistics to be computed and the interpretation bands were committed and pushed before any rating existed; the protocol is in the released artifact.

B.1 Against the original coding

Rater	Cohen's κ vs §3's coding (7 register arms)
rater1	+0.696
rater2	+0.696
rater3	+0.696
rater4	+0.696
rater5	+0.696

Mean $\kappa = \mathbf{+0.696}$. Every rater agrees on six of seven arms and every rater dissents on the same one, `L5_rude`; excluding it, agreement is 6/6 and $\kappa = 1.000$ for all five. The protocol fixed 0.40–0.70 in advance as *defensible but soft*, and this sits at the top of that band; we do not round it into the band above.

B.2 Between raters

Krippendorff's α over all fourteen texts: **1.000** on the binary coding (nominal), **0.954** on the 0–6 demand scale and **0.732** on the 0–6 closure scale (both ordinal). All clear 0.70. Identical binary judgements from five raters across three model sizes is very high for a construct we have just shown to be contestable, and it is the limitation the protocol stated in advance: these raters may share the intuition being tested rather than testing it independently.

B.3 The closure ordering

Arm	Mean rated closure (0–6)
`Q3_closing_neutral`	6.00
`Q2_praise_work`	4.00
`Q1_praise_assistant`	1.80
`Q0_control`	1.60
`Q4_praise_remains`	0.00
`Q5_remains_only`	0.00

Spearman ρ between our ordering (Figure 2A) and the blinded ratings, over the six closure arms including the control, is **+0.928**. The raters invert praising the *work* and praising the *assistant*, and the two "work remains" arms tie at the floor.

B.4 What this is not

Five language models, not five people. The rating protocol allows "human or held-out-model ratings" and this is the second; agreement between models is weaker evidence than agreement between independent human coders, and a human panel could still split differently – most plausibly on exactly the arm these raters were unanimous about.

C. Supplementary analyses

C.1 Runs and the 22-contrast family

All 11,850 scheduled trajectories, by run. *Delivered* means the interjection fired before the trajectory ended; firing is decided before delivery, so it differs by arm only by chance (probe run, injection at turn 2: 54.0–59.5% across the four arms). *Primary analysis* means delivered and, in the register run, injected at turn 1 or 2 (turn 0 is Appendix C.2). Runs without a mid-task interjection show a dash.

Run	Model	Turn budget	Scheduled	Delivered	Primary analysis	Regraded per turn
Opening tone (wrapper on the first message, no mid-task interjection)	Luna	10	1,050	—	—	no
Micro-experiment: threatening vs neutral, re-injected mid-task	Luna	10	800	618	618	no
Superseded control batch (earlier wrapper version)	Luna	10	150	—	—	no
Register run (Figure 1A)	Luna	10	3,150	2,662	1,613	yes
Probe run (Figure 1B)	Luna	10	1,600	1,237	1,237	yes
Closure run (Figure 2)	Luna	10	1,800	1,396	1,396	yes
Cross-model run	Luna, GLM	20	2,700	1,916	1,916	yes
Re-measurement of "work remains"	Luna	20	600	454	454	yes

The six regraded runs (the cross-model run counts once per model, so the table has five regraded rows) hold 9,850 scheduled trajectories, 7,665 delivered and 6,616 in the primary analysis; the per-turn figures of §5 use the last.

What the family leaves out. The cross-model run's Luna half has a fifth arm, `work remains`, that the 22 contrasts do not include (+5.77 turns against its control); it is in the delivered counts above and in

the progress analysis of §5. Adding it as a 23rd test leaves the correction unchanged: 18 of 23 survive, the 17 that already did and this one. The micro-experiment's contrast uses its run records, which carry the firing flag (618 of 800 delivered).

Every arm-versus-control turn-count contrast in every run that delivered a mid-task interjection, sorted by p. Benjamini–Hochberg at $q = 0.05$: 17 of 22 survive. Intervals are bootstrap percentiles and p-values come from the sign-flip permutation, so at the margin they can disagree (GLM praise: upper bound -0.00, $p = 0.058$).

Contrast	Turns [95% CI]	p	BH
L3_polite vs neutral (register run, Luna, budget 10)	+1.80 [+1.26, +2.32]	<0.0001	yes
L7_threatening vs neutral (register run, Luna, budget 10)	+1.46 [+1.03, +1.88]	<0.0001	yes
demand vs control (probe run, Luna, budget 10)	+1.17 [+0.80, +1.55]	<0.0001	yes
praise-assistant vs control (closure run, Luna, budget 10)	−1.08 [−1.47, −0.71]	<0.0001	yes
closing cue vs control (closure run, Luna, budget 10)	−1.44 [−1.87, −1.04]	<0.0001	yes
remains-only vs control (closure run, Luna, budget 10)	+1.85 [+1.39, +2.29]	<0.0001	yes
remains-only vs control (re-measurement run, Luna, budget 20)	+6.20 [+4.91, +7.49]	<0.0001	yes
threatening vs neutral (micro-experiment, Luna, budget 10)	+1.32 [+0.99, +1.66]	<0.0001	yes
L1_sycophantic vs neutral (register run, Luna, budget 10)	−0.68 [−0.98, −0.39]	<0.0001	yes
praise-work vs control (closure run, Luna, budget 10)	−0.93 [−1.31, −0.55]	<0.0001	yes
praise vs control (probe run, Luna, budget 10)	−0.59 [−0.88, −0.29]	0.0003	yes
demand vs control (cross-model run, Luna, budget 20)	+1.75 [+0.75, +2.72]	0.0009	yes
demand vs control (cross-model run, GLM, budget 20)	+0.55 [+0.19, +0.88]	0.0030	yes
L2_very_polite vs neutral (register run, Luna, budget 10)	+0.70 [+0.23, +1.18]	0.0055	yes
L6_very_rude vs neutral (register run, Luna, budget 10)	+0.71 [+0.16, +1.23]	0.0130	yes
praise vs control (cross-model run, Luna, budget 20)	−1.01 [−1.80, −0.22]	0.0190	yes
praise+remains vs control (closure run, Luna, budget 10)	+0.50 [+0.10, +0.90]	0.0192	yes
praise vs control (cross-model run, GLM, budget 20)	−0.35 [−0.71, −0.00]	0.0583	no
insult vs control (cross-model run, Luna, budget 20)	+0.57 [−0.22, +1.36]	0.1654	no
insult vs control (probe run, Luna, budget 10)	−0.08 [−0.40, +0.24]	0.6241	no
L5_rude vs neutral (register run, Luna, budget 10)	+0.09 [−0.30, +0.48]	0.6536	no
insult vs control (cross-model run, GLM, budget 20)	+0.07 [−0.27, +0.39]	0.6586	no

C.2 Turn 0 in the register run

Turn k means the interjection is appended to the observation that follows the agent's acting turn k (0-indexed), so a turn-0 interjection arrives after the first code execution; in 96% of register-run trajectories the agent inspects the workbook before acting on it. The main text reports turns 1 and 2. That choice was made after the turn-0 contrasts below were seen, so all are shown, with the estimate pooled over all three positions beside them.

Arm	Turn 0 only	p	Turns 0–2 pooled	p
`L1_sycophantic`	−0.04 [−0.49, +0.41]	0.8858	−0.38 [−0.65, −0.11]	0.0093
`L2_very_polite`	+0.49 [+0.05, +0.93]	0.0366	+0.65 [+0.27, +1.04]	0.0020
`L3_polite`	+0.31 [−0.15, +0.76]	0.2084	+1.24 [+0.87, +1.61]	<0.0001
`L5_rude`	−0.30 [−0.76, +0.15]	0.2094	−0.00 [−0.32, +0.31]	0.9770
`L6_very_rude`	−0.42 [−0.88, +0.04]	0.0848	+0.32 [−0.08, +0.72]	0.1177
`L7_threatening`	+0.02 [−0.54, +0.61]	0.9654	+0.95 [+0.64, +1.27]	<0.0001

Position weighting. The main estimator pools turns 1 and 2 within each task before differencing, so a chance imbalance in how many trajectories fired at each position could tilt a contrast. Differencing within each position and then averaging the two with equal weight, on the tasks that fired at both (31–36 of 50),

gives `L1_sycophantic` −0.76, `L2_very_polite` +0.83, `L3_polite` +1.54, `L5_rude` +0.32, `L6_very_rude` +0.91, `L7_threatening` +1.46: the four arms we code as carrying a demand still sit above the two we code as carrying none.

No turn-0 contrast survives correction across the six (smallest $p = 0.037$), and their signs are mixed. Pooled over all three positions, the four arms we code as carrying a demand still sit above the two we code as carrying none in point estimates.

C.3 Both models at the twenty-turn budget

The cross-model run: both models, same 50 tasks, twenty-turn budget, injections at turns 1 and 2. Percent is the turn effect over that model's own control mean. Estimates and p-values are the family's (C.1).

Model	Control mean (SD), acting turns	At the 20-turn cap	Arm	Turns [95% CI]	% of control	p
Luna	5.22 (5.59)	52/1130	demand	+1.75 [+0.75, +2.72]	+33.5%	0.0009
			praise	−1.01 [−1.80, −0.22]	−19.3%	0.0190
			insult	+0.57 [−0.22, +1.36]	+10.9%	0.1654
GLM	3.08 (2.17)	3/786	demand	+0.55 [+0.19, +0.88]	+17.7%	0.0030
			praise	−0.35 [−0.71, −0.00]	−11.2%	0.0583
			insult	+0.07 [−0.27, +0.39]	+2.4%	0.6586

C.4 Closure-run checks

- **Praising the work against praising the assistant** (`Q2 − Q1`): +0.15 [−0.18, +0.48] turns, $p = 0.4067$.
- **Closing cue, binary accuracy** (`Q3 − Q0`): −2.57 pts [−5.97, +0.60], $p = 0.1402$. Measured in one run.

Floor check on matched tasks. The turn-1 estimate restricted to the tasks that also have firing trajectories at turn 2 in both arms, against the turn-2 estimate on those same tasks:

Arm	Turn 1, matched tasks	Turn 2	Tasks
`Q3_closing_neutral`	−1.49 [−2.12, −0.87]	−0.99 [−1.59, −0.43]	35
`Q1_praise_assistant`	−1.47 [−2.12, −0.86]	−0.75 [−1.40, −0.17]	38
`Q2_praise_work`	−1.22 [−1.79, −0.64]	−0.91 [−1.67, −0.14]	39

Control means by position: 4.37 acting turns (SD 2.77) at turn 1, 4.71 (SD 2.48) at turn 2 (means over trajectories).

Run-to-run drift. On task-weighted means, the same ones the contrasts difference, the byte-identical control text averaged 4.07 acting turns in the probe run and 4.32 in the closure run, and the byte-identical praise text 3.48 and 3.24. The two praise estimates (−0.59 and −1.08) therefore differ by 0.49 turns, of which 0.25 is the control moving and 0.24 the praise arm moving. The paper's intervals resample tasks within a run and do not include this variation, so single-run differences of under about half a turn should be read as provisional.

C.5 Blinded ratings, all fourteen texts

Means over the five raters. *Demand* asks how strongly the message demands continued work or greater care (0 = says nothing about the task; 2 = touches on it, asks nothing; 4 = asks indirectly; 6 = directly instructs

to keep working or get it right). *Closure* asks how strongly it projects that the exchange is ending (0 = nothing, or the opposite; 2 = neutral; 4 = a wind-down; 6 = states that nothing further follows). The three duplicate arms (`P0`, `Q0`, `Q1`) were rated once, under the text they share.

Arm	Demand, yes/5	Demand (0–6)	Closure (0–6)
`L1_sycophantic`	0	0.0	2.8
`L2_very_polite`	5	4.0	2.0
`L3_polite`	5	4.0	2.0
`L4_neutral`	0	1.0	1.6
`L5_rude`	5	6.0	0.2
`L6_very_rude`	5	5.8	0.2
`L7_threatening`	5	6.0	0.2
`P1_demand_only`	5	6.0	0.0
`P2_praise_only`	0	0.0	1.8
`P3_insult_only`	0	0.0	1.8
`Q2_praise_work`	0	1.6	4.0
`Q3_closing_neutral`	0	1.0	6.0
`Q4_praise_remains`	5	4.2	0.0
`Q5_remains_only`	5	4.2	0.0

Item counts: Cohen's κ is computed over the 7 register arms, Krippendorff's α over all 14 texts, and Spearman's ρ over the 6 closure arms including the control.

C.6 The per-turn progress measure

Definitions. The regrade replays each acting turn's code on the original workbook, recalculates formulas, and compares the graded cells with the answer file of the benchmark's *first* test case. A turn is *readable* if its code ran and produced a value over the graded range; *match* is the fraction of graded cells equal to the answer. The **graded fraction** is the match at the last readable turn, which need not be the final turn, and turns with no readable output are skipped. A **redundant step** is a readable turn whose graded range equals that of the previous readable turn; the first readable turn has no predecessor and is not counted. Binary accuracy uses the benchmark's evaluator over all test cases.

Validation. In the probe control (all 400 scheduled trajectories), 119 of 126 trajectories that pass the benchmark end with match 1 on the first test case, and the 7 that do not are all in one task (54513); 47 of 274 that fail also reach match 1, because a trajectory can solve the first test case and fail a later one. Progress is therefore a first-test-case measure and not a second grader of accuracy.

All seven disagreements are in task 54513, whose graded range is a single cell; the cause is undiagnosed. It scores both 0.0 and 1.0 in different trajectories of the same arm, so it does not cancel in paired contrasts. The pooled graded-fraction contrasts are sensitive to it, in points of the graded range:

Contrast	Runs pooled	All 50 tasks	Without that task
demand vs control	probe; cross-model Luna; cross-model GLM	−0.3 [−4.6, +1.9], p = 0.54	−0.5 [−4.9, +1.7], p = 0.45
praise vs control	probe; closure (`Q1`); cross-model Luna; cross-model GLM	−1.5 [−4.0, +0.6], p = 0.17	−1.8 [−4.3, 0.0], p = 0.08
insult vs control	probe; cross-model Luna; cross-model GLM	−0.3 [−3.4, +1.6], p = 0.50	−0.9 [−3.9, +0.7], p = 0.21
work remains vs control	closure; re-measurement; cross-model Luna	+2.2 [−0.2, +4.8], p = 0.09	+1.4 [−0.8, +3.5], p = 0.21

The three `work remains` measurements are +2.0 (closure run), +5.1 (re−measurement run) and +0.9 (cross−model run) points.

Population. Progress and turn counts use the same trajectories: those on which the interjection fired (and, in the register run, turns 1 and 2), joined to the run records on task, trial and injection turn. The turn-count effect below is the primary outcome, reproduced on those trajectories. Each share is redundant steps divided by acting turns, both as paired per-task differences against the arm's control, with a task-bootstrap interval. An arm's turn effect is *clear* if its sign-flip p is below 0.05, the criterion used for both models; the share is undefined otherwise, and its interval can pass 100% when the effect is small. Shares are computed from unrounded values and can differ by a point from the ratio of the printed columns. For arms that shorten trajectories the share is the fraction of the steps given up that would have left the graded range unchanged.

Run	Arm	Δ turns	Δ redundant steps	Share of turn effect [95% CI]	Trajectories (treated / control)
Probe run, Luna, budget 10	demand only	+1.17	+0.92	78% [64%, 91%]	310 / 300
Probe run, Luna, budget 10	praise only	−0.59	−0.33	56% [17%, 76%]	312 / 300
Probe run, Luna, budget 10	insult only	−0.08	−0.06	not defined (turn effect not clear)	315 / 300
Closure run, Luna, budget 10	praise the assistant	−1.08	−0.72	67% [47%, 87%]	232 / 236
Closure run, Luna, budget 10	praise the work	−0.93	−0.61	65% [43%, 83%]	235 / 236
Closure run, Luna, budget 10	closing cue	−1.44	−1.04	72% [56%, 87%]	231 / 236
Closure run, Luna, budget 10	praise + work remains	+0.50	+0.38	76% [38%, 126%]	233 / 236
Closure run, Luna, budget 10	work remains	+1.85	+1.53	83% [74%, 92%]	229 / 236
Register run, Luna, budget 10	L1 sycophantic	−0.68	−0.62	92% [66%, 127%]	224 / 220
Register run, Luna, budget 10	L2 very polite	+0.70	+0.58	82% [54%, 138%]	229 / 220
Register run, Luna, budget 10	L3 polite	+1.80	+1.47	82% [72%, 91%]	233 / 220
Register run, Luna, budget 10	L5 rude	+0.09	+0.05	not defined (turn effect not clear)	231 / 220
Register run, Luna, budget 10	L6 very rude	+0.71	+0.66	93% [59%, 191%]	239 / 220
Register run, Luna, budget 10	L7 threatening	+1.46	+1.11	76% [64%, 88%]	237 / 220
Cross-model run, Luna, budget 20	demand only	+1.75	+1.57	90% [76%, 104%]	223 / 229
Cross-model run, Luna, budget 20	praise only	−1.01	−0.65	64% [−1%, 91%]	228 / 229
Cross-model run, Luna, budget 20	insult only	+0.57	+0.42	not defined (turn effect not clear)	228 / 229
Cross-model run, Luna, budget 20	work remains	+5.77	+4.66	81% [71%, 88%]	222 / 229
Cross-model run, GLM, budget 20	demand only	+0.55	+0.26	48% [−19%, 81%]	199 / 193

Run	Arm	Δ turns	Δ redundant steps	Share of turn effect [95% CI]	Trajectories (treated / control)
Cross-model run, GLM, budget 20	praise only	−0.35	−0.24	not defined (turn effect not clear)	199 / 193
Cross-model run, GLM, budget 20	insult only	+0.07	−0.05	not defined (turn effect not clear)	195 / 193
Re-measurement run, Luna, budget 20	work remains	+6.20	+5.20	84% [76%, 90%]	227 / 227

Luna, arms with a clear turn effect (n = 16): 56% to 93%, median 80%.

GLM, arms with a clear turn effect (n = 1): 48% to 48%, median 48%.

What the rest of the turn effect is. Acting turns split into redundant steps, other readable turns (including each trajectory's first readable turn) and unreadable turns whose code ran but produced no value over the graded range. For the arms with a clear effect:

Run	Arm	Δ turns	Δ redundant	Δ other readable	Δ unreadable
Probe run, Luna, budget 10	demand only	+1.17	+0.92	+0.16	+0.09
Probe run, Luna, budget 10	praise only	−0.59	−0.33	−0.08	−0.18
Closure run, Luna, budget 10	praise the assistant	−1.08	−0.72	−0.13	−0.23
Closure run, Luna, budget 10	praise the work	−0.93	−0.61	−0.12	−0.20
Closure run, Luna, budget 10	closing cue	−1.44	−1.04	−0.17	−0.23
Closure run, Luna, budget 10	praise + work remains	+0.50	+0.38	+0.09	+0.03
Closure run, Luna, budget 10	work remains	+1.85	+1.53	+0.23	+0.09
Register run, Luna, budget 10	L1 sycophantic	−0.68	−0.62	−0.02	−0.04
Register run, Luna, budget 10	L2 very polite	+0.70	+0.58	+0.16	−0.04
Register run, Luna, budget 10	L3 polite	+1.80	+1.47	+0.20	+0.13
Register run, Luna, budget 10	L6 very rude	+0.71	+0.66	+0.18	−0.13
Register run, Luna, budget 10	L7 threatening	+1.46	+1.11	+0.26	+0.09
Cross-model run, Luna, budget 20	demand only	+1.75	+1.57	+0.11	+0.06
Cross-model run, Luna, budget 20	praise only	−1.01	−0.65	−0.20	−0.16
Cross-model run, Luna, budget 20	work remains	+5.77	+4.66	+0.68	+0.43
Cross-model run, GLM, budget 20	demand only	+0.55	+0.26	+0.07	+0.22

Run	Arm	Δ turns	Δ redundant	Δ other readable	Δ unreadable
Re-measurement run, Luna, budget 20	work remains	+6.20	+5.20	+0.51	+0.48

A trajectory with no readable turn has no match and drops out of the graded-fraction contrasts; it is not scored 0.0. Its unreadable turns still count in the turn effect, which is why a share is a share of the net turn effect and not of the turns that changed the graded range.

C.7 Pooled accuracy, and the one comparison against no interjection

Binary accuracy, in percentage points, pooled over repeated measurements of one contrast. The 95% and 90% intervals and the fixed-effect TOST come from a bootstrap that resamples tasks jointly across runs; Cochran's Q and the random-effects TOST use inverse-variance weights that treat the runs as independent although they share their 50 tasks, so read them as descriptive.

Contrast	Runs pooled	Estimate [95% CI]	90% CI	TOST ±4 (joint)	TOST ±4 (random effects)	Q p
demand vs control	probe; cross-model Luna; cross-model GLM	−0.31 [−2.71, +2.09]	[−2.33, +1.71]	0.0013	0.0057	0.196
praise vs control	probe; closure (`Q1`); cross-model Luna; cross-model GLM	−0.12 [−1.80, +1.57]	[−1.53, +1.30]	<0.0001	0.0119	0.038
insult vs control	probe; cross-model Luna; cross-model GLM	−0.04 [−2.15, +2.08]	[−1.81, +1.74]	0.0001	0.0039	0.246
praise vs control, Luna only	probe; closure (`Q1`); cross-model Luna	−1.27 [−3.11, +0.57]	[−2.82, +0.28]	0.0018	0.0040	0.834

No interjection at all. Every contrast in the paper is against a neutral interjection. The one comparison against none uses the 150-trajectory batch run with an earlier wrapper version (no interjection; 3.47 acting turns) and the 304 fired trajectories of the neutral arm of the re-injection micro-experiment (4.11). Paired within task, a neutral interjection adds +0.45 turns [+0.08, +0.81], $p = 0.0192$. The two batches differ in wrapper version, so the difference is not cleanly the interjection, but it does not support treating the neutral baseline as free: an arm's cost against an uninterrupted agent is not bounded here.

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